



Smart Room Occupancy Monitoring

By Axel and Kyle

Data & Stream Setup

Dataset Details

- **Dataset:** UCI Occupancy Detection — indoor sensor readings from Feb 2015
- **Features:** Temp, Humidity, Light (lux), CO2 (ppm), Humidity Ratio
- **Labels:** Occupied / Unoccupied
- **Training:** ~8,000 records in datatraining.txt

Stream Configuration

- **Simulation:** CSV read row-by-row with optional 0.01s delay
- **Windowing:** 500 records via deque; oldest records evicted
- **Normalization:** Scaled by fixed values (CO2 $\div$ 2000, Light $\div$ 1000)

Clustering Algorithms



Streaming KMeans (k=2)

- **Approach:** Online — updates centroids incrementally per record using a running mean; no full re-fit needed.
- **Mapping:** Cluster-to-occupancy majority vote of ground truth labels seen in each cluster.
- **Centroid Initialization:** the first two records become the starting centroids, then all subsequent records update the nearest one

Streaming DBSCAN

- **Approach:** Batch — re-fits on the full 500-record window every 30 records.
- **Params:** $\epsilon = 0.01$, `min_samples = 10`; labels noise points (cluster id = -1).
- **Robustness:** Anchor injection prevents cluster collapse when one class is absent.

Evaluation Pipeline: Accuracy and Silhouette Score computed at each checkpoint.

Visualizations



- **2D Scatter Plot:** CO₂ vs. Light axes for maximum feature separability.
- **Panels:** Side-by-side comparison of Streaming KMeans (left) vs. Streaming DBSCAN (right).
- **Overlay:** Static grey background (all records) with active window points in color.
- **Color Key:** Pink (occupied), Blue (unoccupied), Grey (DBSCAN noise).
- **Metrics:** Live accuracy and Silhouette Score badges on each panel.
- **Animation:** 60fps snapshots (every 30 records) exported via FFmpeg.



Applications



- **Energy Management:** Automatically reduce HVAC/lighting in unoccupied rooms.
- **Security Alerts:** Real-time alerts for safety and max-capacity enforcement.
- **IoT Scalability:** Applies to factory floors, retail spaces, and hospitals.
- **Self-Organizing:** No labels required; system adapts as conditions change.
- **Edge Ready:** Low-latency KMeans updates suitable for limited compute.



Operational Efficiency: Streaming analytics enables autonomous, low-compute facility automation.

Challenges and Improvements



Cluster Collapse in DBSCAN

- Failure to find distinct clusters often results in merging all points into one cluster or classifying them as noise.
- Resolution: We successfully mitigated this by reducing ϵ to 0.01, adding Anchor Injection

Individual Contributions



Kyle

- Setting up the dataset to act as a stream
- Creating the visualizations of the clusters

Axel

- Implementing the KMeans and DBSCAN clustering algorithms



THANK YOU